CS 272.01

Exam 1 Review

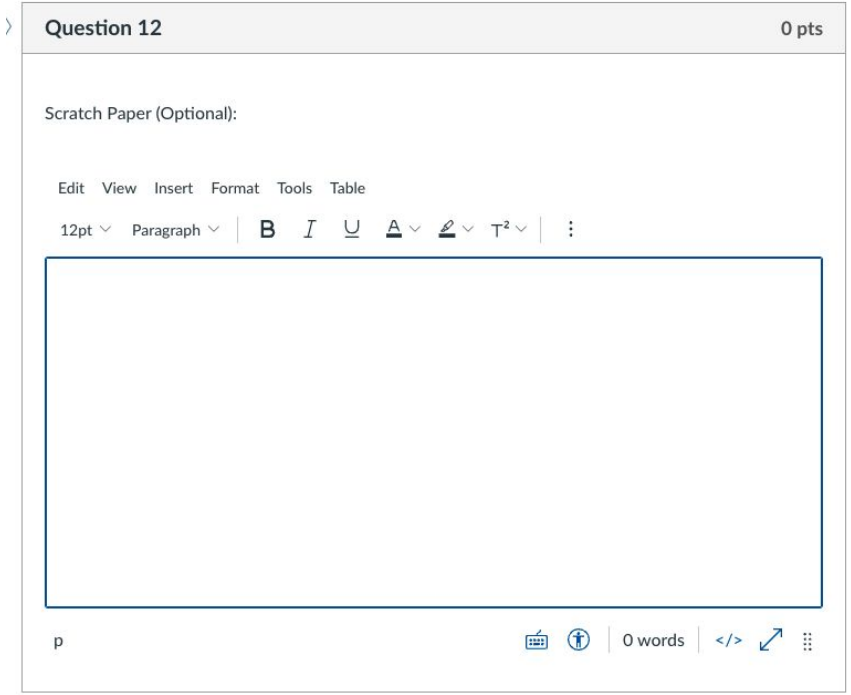
Date/Format

Date: Monday, March 9 - During the class

Format:

- <u>In-person ONLY</u> for proctoring reasons
  - You cannot access the Canvas quiz from your home
  - Canvas exam (Respondus browser required)
- 60 mins
- A single-sheet cheat sheet is allowed (Two-sided; Can be printed)
- Calculator allowed (You need it!)
- Scratch workspace will be provided on Canvas
  - Not graded for the initial grade
  - Leave your work for partial credits, if needed
- Physical scratch paper will be collected but NOT be graded.

Exam Format


Question 12
0 pts
Scratch Paper (Optional):
Edit View Insert Format Tools Table
12pt
Paragraph
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0 words

Materials

- Slides
  - #01 What is learning?
  - #02 Markov Decision Processes
  - #03 Dynamic Programming
  - #04 MAB
  - #05 MC
  - #06 TD
- Coding
  - You should be familiar with all pseudo code and its python implementation
  - PA1
  - DP, MAB, MC, TD

Question Format

- Conceptual questions like Wednesday quizzes
  - Multiple choice
  - T/F
  - Short answer
- Calculation in simple examples like Milestone questions
  - Value functions
  - Policy
- Code related questions
  - Fill in a small part of the code
  - Explain the meaning of a certain part of the PA
  - Read the code and understand what it is

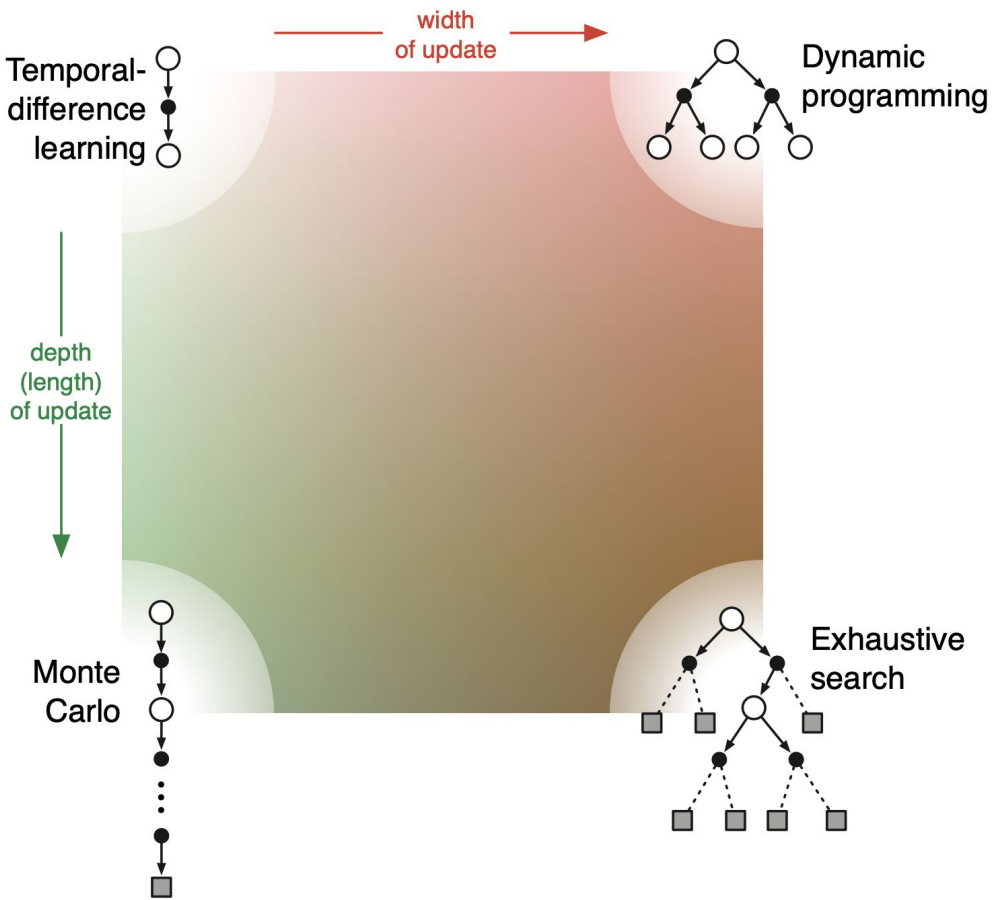
Check points: You should be able to

- Compute the Q value update by different types of model free control algorithms (for a given episode)
  - MC control, SARSA, Q-learning (see the milestone questions)
- Calculate the Q value updates by **Batch** MC and TD(0)
- Know the conceptual differences among the model-free prediction and control algorithms
  - MC, TD, TD(lambda), MC control, SARSA, Q-learning, Double Q-learning
  - Sampling
  - Bootstrapping
  - Bias and Variance
- Understand the different types of policies and when they are used
  - Soft policy vs the greedy policy extraction

Check points: You should be able to

- On-policy vs off-policy
  - What is the importance sampling?
- Compute the state value function v in a Markov Decision Process
  - Practice with an example on lecture slides (Malfunctioning robot)
- Know the definition of an optimal policy and Policy Improvement Theorem
  - Definition slides
- Understand how the epsilon-greedy, UCB, and optimistic initialization work

Summary


width
of update
Temporal-
difference
learning
Dynamic
programming
depth
(length)
of update
Monte
Carlo
Exhaustive
search

Must-Know Formula

$$V(S) \leftarrow V(S) + \alpha\big[R + \gamma V(S') - V(S)\big]$$

Can be a return G too…

$$Q(S, A) \leftarrow Q(S, A) + \alpha\big[R + \gamma \max_a Q(S', a) - Q(S, A)\big]$$

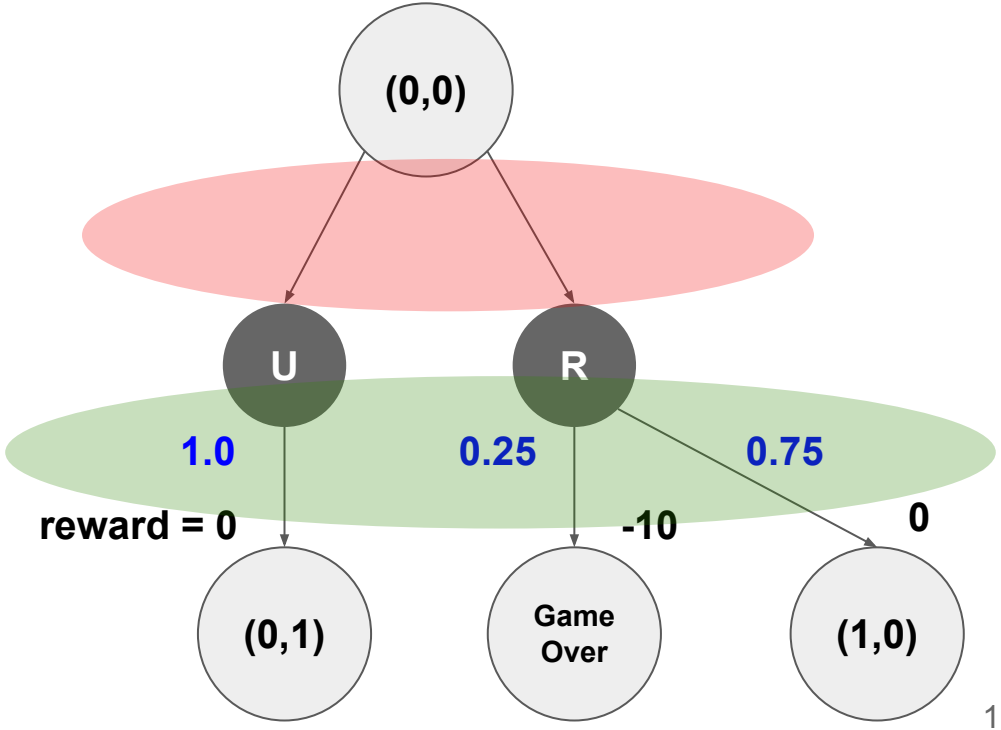
Policy vs Transition Probability

Policy defines which action to take

- Intentional
- You have control over it
- A better policy gives better rewards

Transition decides where to go

- Given in the environment
- You do NOT have control
- A better policy should be selected guessing this probability



Select ALL that an RL agent has control over. (i.e., Thing(s) an RL agent can change.)

- A state space
- A reward function
- A policy function
- An action space
- A transition probability function

Select the most appropriate description of the optimal action value function q*(s,a)

- The agent is assumed to follow the best trajectory after the successor state.
- The agent is assumed to try multiple trajectories from the successor state and use the weighted average to assess the state-action pair.
- Action a in q*(s,a) is always the best action (with the highest reward) from the current state s.

Policies in this category try all possible actions from each state.

- Off-policy
- On-policy
- Greedy policy
- Soft policy

Choose all method(s) that use(s) sampling.

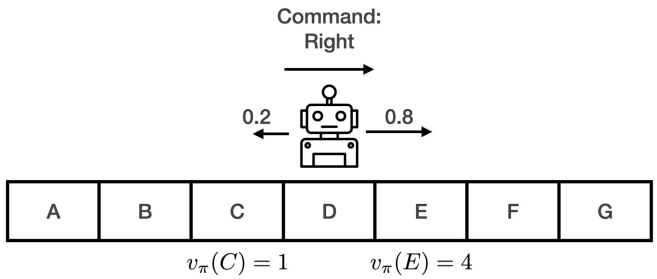
- MC
- Q-learning
- SARSA

- A reward indicates the long-term impact of an agent's action. T/F
- A policy is a function that maps each state to an action (or actions). T/F
- When the discount factor (gamma) is close to 1, the impact of the rewards in the future transitions will be smaller. T/F
- The policy improvement theorem suggests that we are always able to create a new policy that is better or at least equal to the current policy by taking an action with the maximum action value. T/F
- Policy Evaluation updates the policy until the policy converges to an optimal one. T/F
- On-policy MC control must use the greedy policy update. T/F
- Between MC and Sarsa, which one would experience a higher variance?

Milestone Questions

- A robot is in a cell (State)
- I can move the robot to right or left (Two possible actions)
- Now, from my experience, I decided to choose RIGHT in cell D 60 % of the time (Policy)
- BUT the robot is malfunctioning and moves to the opposite direction with 20% (Transition probability)

$$q_\pi(D, \text{right}) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')$$



$v_\pi(C) = 1$ $v_\pi(E) = 4$

$R_D^{right} = 3$

$R_D^{left} = 3$

Milestone Questions

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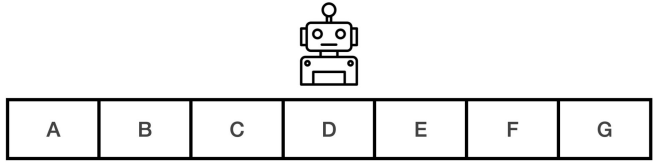
$$v_\pi(D) = \sum_{a \in \mathcal{A}} \pi(a \mid s) q_\pi(s, a)$$



Milestone Questions

- From the current state s,
- There are five actions a1, a2, …, a5
- Current estimate of q-values are:
  - Q(s,a1) = 0.2
  - Q(s,a2) = 1.0
  - Q(s,a3) = 0.8
  - Q(s,a4) = 0.2
  - Q(s,a5) = 0
- Compute the probability that each action will be selected during the next visit to s under the epsilon-greedy algorithm (epsilon = 0.1)

Milestone Questions

- From the current state s,
- There are five actions a1, a2, …, a5
- Current estimate of q-values are:
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  - Q(s,a3) = 0.8
  - Q(s,a4) = 0.2
  - Q(s,a5) = 0
- If a1, a2, a3, and a4 have been tried 1000 times and a5 has been tried only 1 time, which action will be selected in the next round by the UCB?

$$a_t = \underset{a \in \mathcal{A}}{\operatorname{argmax}} \left(\bar{Q}(a) + \sqrt{\frac{2 \ln t}{N_t(a)}} \right)$$

Milestone Questions

- There are two actions (Left or Right) from each state
- The current Q table shows
  - Q(S1, L) = 1.0, Q(S1, R) = 0
  - Q(S2, L) = - 1.0, Q(S2, R) = 2.0
  - Q(S3, L) = 5.0, Q(S3, R) = - 1.0
- Assume an agent observed the following episode
  - S1 L 3 S2 R 4 S3 R 10 T
  - This episode was generated, following the epsilon-greedy
- Using On-policy first-visit MC control,
  - Compute the action value (assume no discounting for simplicity)
  - Update the policy function

Milestone Questions: Batch Updates

- Assume:
  - Two states A, B
  - No discounting
  - The following eight episodes will be sampled again and again
    - A 0 B 0
    - B 1
    - B 1
    - B 0
- Compute V(A) and V(B) with MC and TD(0)

Milestone Questions: Batch Updates

- MC
  - V(A) = 0
  - V(B) = 1 / 2
- TD(0)
  - V(A) = 1 / 2 (Bootstrapping)
  - V(B) = 1 / 2

Milestone Questions

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  - Q(S3, L) = 5.0, Q(S3, R) = - 1.0
- Assume an agent observed the following experiences
  - S1 L 3 S2
  - S2 R 4 S3
- This episode was generated, following the epsilon-greedy
- Using Q-learning,
  - Compute the action value (assume no discounting for simplicity)
  - Update the policy function